

6	(a)	A metal plate has work function of 4.5 eV. When ultraviolet radiation of wavelength 120 nm is incident on the plate, a photoelectric current is detected.
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		(i)	Calculate the maximum speed of the photoelectrons as a result of this illumination.	
			maximum speed = m s ⁻¹	[2]
			Using the Einstein photoelectric equation, $hf = \phi + KE_{\max}$ $hf = \phi + \frac{1}{2} m_e v_{\max}^2$ $v_{\max} = \sqrt{\frac{2 \left(h \frac{c}{\lambda} - \phi \right)}{m_e}}$ $= \sqrt{\frac{2 \left(6.63 \times 10^{-34} \times \frac{3.0 \times 10^8}{120 \times 10^{-9}} - 4.5 \times 1.6 \times 10^{-19} \right)}{9.11 \times 10^{-31}}}$ $= 1.43 \times 10^6 \text{ m s}^{-1}$	
		(ii)	The uncertainty in the speed of the photoelectrons in (a)(i) is 0.50%. Assuming that the photoelectrons are moving in the horizontal direction,	
		1.	determine the uncertainty in the horizontal momentum of the photoelectrons;	
			uncertainty = N s	[3]
			Momentum of the electron: $p_x = m_e v_x$ $= 9.11 \times 10^{-31} \times 1.43 \times 10^6$ $= 1.30 \times 10^{-24} \text{ kg m s}^{-1}$ $\frac{\Delta p_x}{p_x} = \frac{\Delta v_x}{v_x} = \frac{0.50}{100} \text{ since mass of electron is constant}$ $\frac{\Delta p_x}{p_x} = \frac{0.50}{100}$ $\Delta p_x = \frac{0.50}{100} \times 1.30 \times 10^{-24}$ $= 0.70 \times 10^{-26} \text{ kg m s}^{-1} \text{ (must be 2 s.f. Do NOT accept 1 s.f.)}$	
		2.	determine the minimum uncertainty in the position of the electron.	
			uncertainty = m	[2]
			Applying Heisenberg's uncertainty principle, $\Delta p_x \Delta x \gtrsim h$ $\Delta x \gtrsim \frac{h}{\Delta p_x}$ $\Delta x \gtrsim \frac{6.63 \times 10^{-34}}{0.70 \times 10^{-26}}$	

$$= 9.5 \times 10^{-8} \text{ m}$$

Also accept if candidate uses the exact form of Heisenberg's uncertainty principle $\Delta p_x \Delta x \geq \frac{h}{4\pi}$

(b) Fig. 6.1 shows an x-ray spectrum.

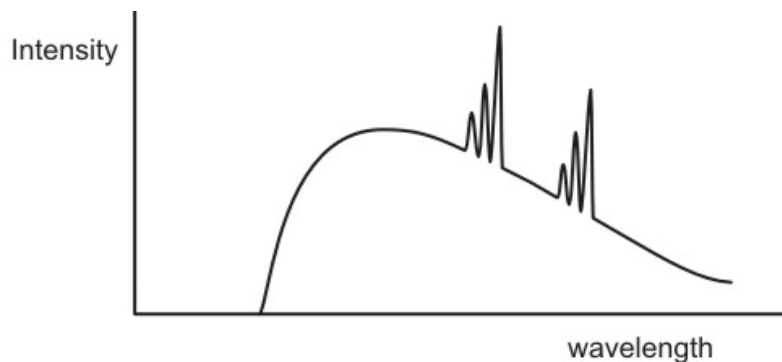


Fig. 6.1

Describe how the continuous spectrum is produced.

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[4]

A high speed electron interacts with the nucleus of a target metal. It experiences a net force due to the electric force exerted by the target nucleus, and thus **accelerates/decelerates**. This **produces an x-ray photon of energy equal to the kinetic energy lost by the incident electron** via Bremsstrahlung process.

If an electron has excess kinetic energy after the deceleration, it will continue to interact with another nucleus in the metal, and undergo another Bremsstrahlung process / further deceleration, emitting another photon of a different wavelength.

This process continues until the incident electrons completely loses its kinetic energy. **This series of Bremsstrahlung processes / series of decelerations produce photons of a (continuous) range of wavelengths.**